

PAPER The Aegis Forward-Base Burner: Q_E Closure, Plug Requirement, and Mirror Confinement in an Islanded Low-Neutron D– ^3He Tandem Mirror

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Abstract

Aegis is the forward-base configuration of the Kronos deuterium–helium-3 tandem-mirror burner: a fixed, hardened, grid-independent installation that turns an aneutronic-dominated fusion source into firm electrical power for a defence site without a fuel-convoy tail. We present the integrated physics case for that machine and separate, throughout, what is robust from what is conditional. At the frozen design point the plant reaches an engineering gain $Q_E = 1.318$ at fusion power 4298.5 MW, of which 3868.0 MW is the charged-particle $\text{D} + ^3\text{He} \rightarrow p(14.68\text{ MeV}) + \alpha$ channel and only 233.95 MW is neutron power — a neutron fraction $f_n = 5.44\%$, the low-activation advantage that motivates the fuel and the siting. The binding physics gate is the end-plug potential: confining the central-cell ions requires a plug electrostatic potential $\phi_i = 248.75\text{ keV}$, which through the ambipolar Boltzmann relation demands a plug-to-central density ratio $n_p/n_c = \exp(e\phi_i/T_e) = 16$ at carrier temperature $T_e = 89.7\text{ keV}$ — a requirement more than two orders of magnitude beyond the demonstrated tandem-mirror record, retired only by a dedicated plug-potential experiment. A Saltelli–Sobol analysis confirms the plug ratio is the first-order driver of Q_E (first-order index 0.41, total-order 0.87), and a 40 000-sample Monte-Carlo scan returns $P(Q_E > 1) = 1.0$ across the design ranges. We are explicit about a second binding lever: direct energy conversion is a declared, load-bearing input on a conservative-to-ideal ladder ($\eta_{\text{dec}} = 0.49$ conservative canon, 0.70 design basis, 0.85 proton-only ceiling), and net electric requires η_{dec} above a break-even near 0.52, so the design point closes at the design-basis efficiency and the conservative floor does not — a lever, not a free upside. The loss-cone microstability sits on the drift-cyclotron-loss-cone knee at a 6.25 % warm-plasma fill, a threshold rather than a margin, quenched in particle-in-cell modelling by a maintained energetic-stabiliser fraction. The Aegis product wrap is stated quantitatively: a black-start whose direct-converter output crosses the start-up draw at $\approx 42\text{ s}$ on $\sim 3\text{ MWh}$ of stored energy, an islanded steady state at $+104\text{ MWe}$, a fleet capacity factor of 68–96 % across redundancy tiers, and a one-year grid-independent fuel package of order one tonne. Every net-power number is contingent on retiring the plug gate; the neutron-source performance and the low-activation siting case are not.

Keywords: tandem mirror; deuterium–helium-3; plug potential; loss-cone microstability; direct energy conversion; forward-base power; islanded microgrid; low-neutron fusion; advanced fuel

Contents

Nomenclature	3
1 Introduction	3
1.1 The forward-base power problem and the Aegis configuration	3
1.2 Why a tandem mirror	3
1.3 Contribution and separation of claims	3
2 Governing equations and the formal problem	4
2.1 Reaction rate, fusion power and the neutron fraction	4
2.2 Ambipolar confinement and the plug-potential requirement	4
2.3 The recirculation-limited engineering gain	5
2.4 Direct recovery of the axial stream	5
2.5 Loss-cone microstability	5
2.6 Synchrotron reabsorption pins the electron temperature	5
3 Computational methodology	6
3.1 Zero-dimensional balance and design-space scan	6
3.2 Kinetic and microstability modelling	6
3.3 Neutronics, dose and structural qualification	6
4 Results	6
4.1 The design point	6
4.2 The fuel choice	6
4.3 The binding gate: the plug potential	8
4.4 The second lever: direct-converter efficiency and net-power break-even	8
4.5 Sensitivity and uncertainty	9
4.6 Loss-cone microstability	9
4.7 Forward-base operating envelope	10
4.8 The plug magnet and the fuel logistics	10
4.9 Verification status	11
5 Discussion	11
5.1 Against the tandem-mirror record	11
5.2 Against the direct-conversion and advanced-fuel literature	12
5.3 Why Aegis specifically	12
6 Limitations	12
7 Conclusion	12
Acknowledgments	13
Data availability	13
Competing interests	13
References	14

Nomenclature

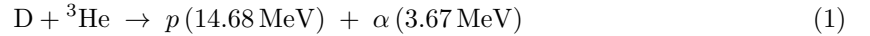
Symbol	Definition
$\bar{Q}_E$	plant engineering gain (gross electric out / recirculating in)
P_{fus}	total fusion power; P_{ch} charged, P_n neutron
f_n	neutron fraction of total fusion power
n_p/n_c	plug-to-central-cell density ratio (the binding gate)
ϕ_i	ion-confining electrostatic plug potential
T_e, T_i	carrier electron / central-cell ion temperature
R_m	mirror ratio $B_{\text{max}}/B_{\text{min}}$
τ_p	Pastukhov (plugged) axial ion-confinement time
f_s	energetic-stabiliser (warm-plasma) fill fraction inside the loss cone
η_{dec}	direct-energy-converter module efficiency (declared input)
f_{dir}	directed (recoverable) fraction of the axial loss stream
γ/ω_{ci}	normalised loss-cone (DCLC/AIC) growth rate
β_c	central-cell plasma beta

1 Introduction

1.1 The forward-base power problem and the Aegis configuration

A fixed forward defence installation is a demanding electrical load: it must run through a grid outage, it must not depend on a fuel convoy that is itself a target, and it must accept a directed-energy or high-power-radar duty that swings by tens of megawatts on a timescale the prime mover has to follow. Diesel and gas turbines answer the first requirement and fail the second; a fission microreactor answers both but carries a spent-fuel and proliferation signature that constrains where it may be sited. Aegis is the Kronos answer: the forward-base housing of a deuterium–helium-3 ($\text{D}-^3\text{He}$) tandem-mirror burner, a fixed, hardened installation whose fuel is a sub-cubic-metre annual charge and whose neutron output is a small fraction of a same-power deuterium–tritium ($\text{D}-\text{T}$) machine. Aegis and its sibling data-centre configuration, MetroVolt, are the *same* burner core in two housings; this paper treats the physics of the core and the forward-base product wrap, and its companion treats the data-centre wrap [33].

The attraction of the fuel is aneutronicity, and its difficulty is confinement. The primary reaction



releases all of its energy in charged particles, so a device that burns it produces neutrons only from the unavoidable $\text{D}-\text{D}$ side reactions and can, in principle, recover the charged-particle power directly rather than through a steam cycle. But the reaction peaks near $T_i \sim 60\text{--}100 \text{ keV}$, an order of magnitude hotter than a $\text{D}-\text{T}$ plasma, and at those temperatures the electrostatic well that a tandem mirror uses to plug its ends must be correspondingly deep. The central thesis of this paper is that the resulting *plug-potential requirement* is the single binding physics gate of the machine, and that an honest forward-base case names it as such rather than hiding it behind an engineering-gain headline.

1.2 Why a tandem mirror

The tandem-mirror concept [1, 2] confines a long, low-field central cell between two high-field end plugs that hold an electrostatic potential ϕ_i ; the potential, not a magnetic mirror alone, is what confines the central-cell ions against axial loss. The concept was reduced to practice on GAMMA-10, TARA and MFTF-B, where thermal-barrier and plug potentials were formed and measured [6, 5], and it has returned to prominence with high-temperature-superconductor axisymmetric mirrors such as WHAM, whose high mirror field is exactly the enabling technology a compact $\text{D}-^3\text{He}$ plug needs [8, 7]. A mirror is attractive for an advanced fuel for three reasons that a closed toroidal device cannot match: it is intrinsically steady-state and disruption-free; its open ends expose a directed particle stream that a direct-energy converter can decelerate; and its linear geometry modularises for transport and field assembly [29]. The price is the plug, and the plug is the subject of most of what follows.

1.3 Contribution and separation of claims

This paper contributes the integrated physics case for the Aegis forward-base burner: the governing power balance and its formal problem statement (§2); a GPU-accelerated computational methodology naming the real codes (§3); a results section with the design point, the fuel choice, the plug gate and its sensitivity, the recirculation-limited gain, the loss-cone microstability, and the forward-base operating envelope — black-start, islanded load-following and availability —

grounded in eleven figures and four tables (§4); a comparison against the published tandem-mirror, direct-conversion and advanced-fuel literature (§5); a short limitations section (§6); and a conclusion. Throughout, every headline number is a frozen anchor in the Kronos Physics De-Risking Register [30], and each carries its serial (e.g. BU-L1-A1, AG-L1-A1) so it is auditable. We separate, in every section, the *robust* neutron-source and low-activation performance — which does not depend on the plug — from the *conditional* net-power performance, which does.

2 Governing equations and the formal problem

2.1 Reaction rate, fusion power and the neutron fraction

For a two-species reacting plasma of number densities n_1, n_2 the volumetric fusion power in channel r is

$$p_r = \frac{n_1 n_2}{1 + \delta_{12}} \langle \sigma v \rangle_r(T_i) E_r, \quad (2)$$

with δ_{12} the Kronecker symbol removing double counting for like species and $\langle \sigma v \rangle_r$ the Maxwellian-averaged reactivity, which we evaluate with the Bosch–Hale parametrisation [9]. The central-cell burn runs the primary channel (1) plus the two D–D branches,

$$\text{D} + \text{D} \rightarrow \text{T} (1.01 \text{ MeV}) + p (3.02 \text{ MeV}), \quad (3)$$

$$\text{D} + \text{D} \rightarrow {}^3\text{He} (0.82 \text{ MeV}) + n (2.45 \text{ MeV}), \quad (4)$$

and a small secondary D–T burn on the bred triton. Integrating (2) over the central-cell volume gives the total fusion power and its split into a charged part P_{ch} and a neutron part P_n ; the neutron fraction

$$f_n = \frac{P_n}{P_{\text{fus}}} = \frac{P_n}{P_{\text{ch}} + P_n} \quad (5)$$

is set almost entirely by the D–D neutron branch and the secondary D–T burn, because the primary channel is aneutronic. At the design point $P_{\text{fus}} = 4298.5 \text{ MW}$, $P_{\text{ch}} = 3868.0 \text{ MW}$ and $P_n = 233.95 \text{ MW}$, so $f_n = 5.44\%$ (gate BU-L1-A1) — roughly a factor of fifteen below the $\approx 80\%$ neutron fraction of a D–T plasma of the same fusion power, which is the whole activation-and-siting argument for the fuel and for a forward-base machine.

2.2 Ambipolar confinement and the plug-potential requirement

The central-cell ions are not confined by the magnetic mirror alone — the mirror ratio R_m leaves an unavoidable loss cone of half-angle $\theta_L = \arcsin R_m^{-1/2}$ — but by an electrostatic potential ϕ_i raised in the end plugs. In the plug the electrons obey a Boltzmann relation, so the density needed to build a confining potential ϕ_i at carrier temperature T_e is

$$\boxed{\frac{n_p}{n_c} = \exp\left(\frac{e\phi_i}{T_e}\right)} \quad (6)$$

(gate BU-L2-A5). Because the requirement sits inside the exponent, the carrier electron temperature is the strongest single lever on it. At $\phi_i = 248.75 \text{ keV}$ and $T_e = 89.7 \text{ keV}$ equation (6) gives $n_p/n_c = \exp(248.748/89.717) = 16$ exactly. This is a requirement-class target: the plug potential is more than two orders of magnitude beyond the demonstrated tandem-mirror record [6], and no design study can retire it — only a dedicated plug-potential experiment at reactor-relevant scale can.

The rate at which plugged central-cell ions leak through the ambipolar barrier is the Pastukhov problem [3, 4]. For an electrostatically plugged mirror the parallel ion-confinement time scales as

$$\tau_p \simeq \tau_{ii} \frac{\sqrt{\pi}}{2} g(R_m) \left(\frac{e\phi_i}{T_i}\right) \exp\left(\frac{e\phi_i}{T_i}\right), \quad (7)$$

with τ_{ii} the ion–ion collision time and $g(R_m)$ a weak logarithmic function of the mirror ratio; the exponential dependence on $e\phi_i/T_i$ is what makes a deep well so valuable and its formation so demanding. At the design point the model returns $\tau_p = 3.73 \text{ s}$ with $e\phi_i/T_i = 2.764$ and mirror ratio $R_m = 4.61$ (gate BU-L1-A1). The central-cell ions are well magnetised, with a device-to-gyroradius ratio $a/\rho_i = 143.9$ and an effective charge $Z_{\text{eff}} = 1.462$; the central-cell beta $\beta_c = 0.55$ enters the balance as an assumed input rather than an MHD-solved result, and is carried as such (§6).

2.3 The recirculation-limited engineering gain

The plant engineering gain is not gated by the fusion gain but by the power that must be recirculated to sustain the plug and the thermal barrier. Writing η_{th} for the thermal-cycle efficiency on the residual heat, η_{dec} for the direct-converter module efficiency on the recovered charged stream P_{ch} , and $P_{\text{plug}}, P_{\text{aux}}$ for the recirculating plug and auxiliary powers,

$$Q_E = \frac{\eta_{\text{th}} (P_{\text{fus}} + P_{\text{aux}}) + \eta_{\text{dec}} P_{\text{ch}}}{P_{\text{plug}} + P_{\text{aux}}} \quad (8)$$

(gates BU-L1-A1, BU-L1-A10). Two features of (8) organise the entire design. First, Q_E is bounded from below by P_{plug} : a deeper plug improves confinement through (7) but costs recirculating power, and the optimum is a balance, not a limit. Second, Q_E is *linear* in η_{dec} , so the converter efficiency is a direct multiplier on the recovered charged power and therefore a load-bearing lever, treated in §4.4 and in detail in the direct-conversion companion [35]. At the design point equation (8) returns $Q_E = 1.318$ with the converter at its design-basis efficiency.

2.4 Direct recovery of the axial stream

The open ends of a mirror expose a directed particle stream carrying an axial power $P_{\text{ax}} = 2719.7 \text{ MW}$ at the design point. A direct-energy converter decelerates the charged component against a graded electrostatic potential and collects it as DC current at efficiency η_{dec} ; the recovered electric power is

$$P_{\text{dec}} = \eta_{\text{dec}} f_{\text{dir}} P_{\text{ax}}, \quad (9)$$

with $f_{\text{dir}} = 0.495$ the directed (recoverable) fraction of the stream (gate BU-L1-A10). At the design efficiency the converter returns 1903.8 MW_e , or 54.1 % of the plant electric output $P_{\text{out}} = 3522.0 \text{ MW}_e$. Because P_{dec} scales directly with η_{dec} , the net electric output $P_{\text{net}} = P_{\text{out}} - P_{\text{recirc}}$ crosses zero at a break-even efficiency near $\eta_{\text{dec}} \approx 0.52$; below it the plant does not reach net electric. This break-even is the second binding lever of the design and is made quantitative in §4.4. A physics subtlety relocates the converter problem: the dominant axial channel is a directed *electron* stream rather than the often-assumed directed-ion stream, which changes the converter geometry and is the subject of the direct-conversion companion [35].

2.5 Loss-cone microstability

The plug's loss-cone distribution is a free-energy source for the drift-cyclotron-loss-cone (DCLC) and Alfvén-ion-cyclotron (AIC) instabilities [5]. Their marginal stability is governed by the balance between the loss-cone drive and the stabilising influence of a warm (isotropic) plasma component of fill fraction f_s ; schematically the normalised growth rate obeys

$$\frac{\gamma}{\omega_{ci}} = \Gamma_{\text{LC}} \left(\frac{\partial f}{\partial v_{\perp}} \right) - \Lambda(f_s, \nabla n_{\perp}), \quad (10)$$

with Γ_{LC} the loss-cone drive and Λ the warm-plasma and density-gradient stabilisation. The design sits on the knee: a warm-plasma fill of 6.25 % is the $\gamma = 0$ threshold rather than a comfortable margin (gate BU-L2-A5), and the residual AIC branch is cleared only for sufficiently broad plug-edge gradients. This is a threshold, and we present it as one; the detailed particle-in-cell microstability study is the companion paper [38], whose result we reproduce independently in §4.6.

2.6 Synchrotron reabsorption pins the electron temperature

Because T_e enters (6) exponentially, the value of T_e at the operating point is a physics result, not a free choice, and it is set by synchrotron transport. Electron-cyclotron emission at harmonic n has an optical depth τ_n that falls with n ; the plasma re-absorbs as a black body up to a cutoff harmonic n^* where $\tau_{n^*} \sim 1$ and radiates freely above it, so the reabsorbed loss is

$$P_{\text{sync}} \simeq \int_{n^*}^{\infty} \eta(n) [1 - e^{-\tau_n}] dn, \quad n^* \propto \left(\frac{\omega_{pe}^2}{\omega_{ce}} a \right)^{1/3} \sim a^{1/3}, \quad (11)$$

so the cutoff — and hence the reabsorbed fraction — grows as the one-third power of the minor radius. The maximum- Q_E operating point is the temperature at which the synchrotron-pinned T_e and the plug requirement (6) are jointly satisfied; the full electron-cyclotron transport calculation and its model spread are the synchrotron companion [40], which fixes the maximum- Q_E point at $T_e \approx 89.7 \text{ keV}$. This closes the formal problem: given the fuel and geometry, equations (6)–(11) determine the plug requirement, the confinement, the gain, and the microstability threshold self-consistently.

3 Computational methodology

Every quantitative claim in this paper is produced by an explicit code on an explicit compute resource, and each is re-run independently against a second library before it is frozen into the register [30, 31]. Training of the machine-learning surrogates and the large ensemble scans run on the NVIDIA GPU HPC campaign; the reduced-order solvers, the zero-dimensional balance and the analytic monitors run on the in-house HPC node, so the fast design loops never contend with the training fleet. No result below depends on unpublished or proprietary data.

3.1 Zero-dimensional balance and design-space scan

The power balance of equations (2)–(8) is evaluated by a mirror solver (`solve_mirror`) built on the Bosch–Hale reactivities [9] and the Pastukhov end-loss model [3]. The design point (BU-L1-A1) and the net-gain database (BU-L2-A2) are reproduced to the anchor values ($Q_E = 1.3181$, $P_{\text{fus}} = 4298.5$ MW). Global uncertainty is quantified by a 40 000-sample Saltelli design over the input ranges, from which we compute both the Monte-Carlo distribution of Q_E (KX-L1-A5-BU) and the first- and total-order Sobol sensitivity indices (HX-05); the Saltelli estimator [14] gives the variance decomposition that identifies the plug density ratio as the dominant driver.

3.2 Kinetic and microstability modelling

The loss-cone microstability of equation (10) is modelled with electromagnetic particle-in-cell (PIC) simulation. We use the GPU-native code `WarpX` [23] for the one-dimensional hybrid plug runs (BU-L2-A7, AG-CX-03, AG-CX-07) and cross-check selected cases with `SMILEI` [24]; both are run in a deliberately conservative regime and report the normalised growth rate γ/ω_{ci} as a function of warm-plasma fill. Ion-distribution shaping, RF alpha-channeling of the 14.68 MeV proton, and the non-Maxwellian tail that feeds the converter are treated with a one-dimensional Fokker–Planck solver (BU-L2-A8) [37]. The gyrokinetic transport context for the central-cell electrons is provided by the CGYRO/GENE family [25, 26], whose expensive maps are emulated at control latency by neural-operator surrogates [28] in the control companion [36]; the same learned-control line that has been demonstrated in the toroidal case [27] carries over to the burner’s actuator set.

3.3 Neutronics, dose and structural qualification

The neutron source term from the D–D and secondary D–T branches, the personnel-proximity dose, and the shield sizing for the fixed installation are computed with the Monte-Carlo transport code `OpenMC` [22] (AG-L1-A2). The plug coil — the high-field REBCO magnet that is the structural long pole — is qualified by three-dimensional finite-element analysis with `CalculiX` on the as-built CAD (P3-FEA-10); the winding-pack field basis is the conservative 26.49 T FEA value, distinct from the breeder centrepost’s 16.84 T (two separate magnets, never conflated) [34]. Time-domain fleet availability (HX-01) is an event-driven Monte-Carlo, and the black-start trajectory (HX-07) is a reduced ordinary- differential-equation integration on the real D– ^3He reactivity. Each serial in this paper names a run that was reproduced on a second library before freezing.

4 Results

4.1 The design point

Table 1 collects the frozen design point. The machine reaches $Q_E = 1.318$ at 4298.5 MW fusion power with a 5.44 % neutron fraction; the plug potential is 248.75 keV, the mirror ratio 4.61, and the plugged ion confinement time 3.73 s. Figure 1 shows the power flow: the charged channel dominates, the neutron channel is small, and the net islanded output is a modest +104 MW_e because most of the electric output recirculates to the plug.

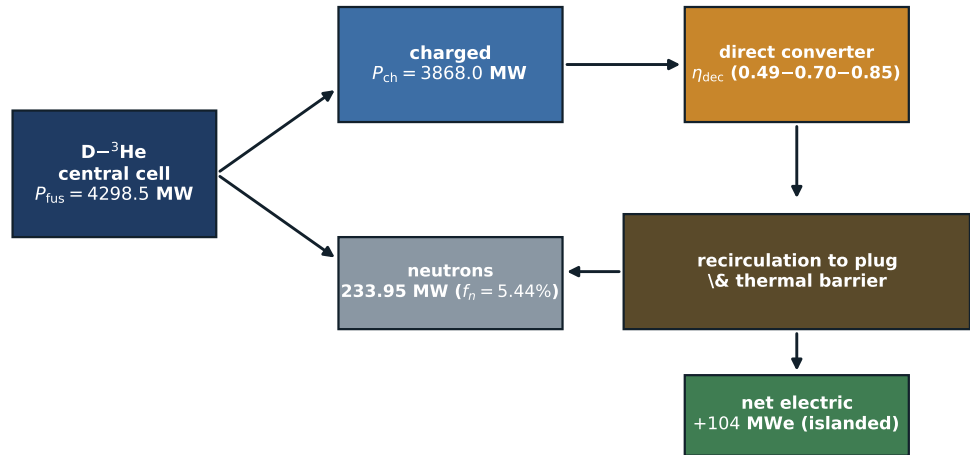
4.2 The fuel choice

The forward-base case rests on the neutron fraction, and the neutron fraction rests on the reactivity ladder of figure 2, computed from the Bosch–Hale parametrisation [9]. D–T peaks near 64 keV and is almost entirely neutronic; D– ^3He peaks an order of magnitude hotter and is aneutronic in its primary channel; catalysed D–D gives no engineering-gain benefit and a worse neutron budget (AG-CX-14). At the burner operating point $T_i = 90$ keV the D– ^3He reactivity is $\approx 1.5 \times 10^{-16} \text{ cm}^3 \text{ s}^{-1}$, and the fuel choice roughly halves the neutron fraction relative to catalysed D–D (5.4 % versus ≈ 9.5 %) while sitting at the engineering-gain optimum. The low neutron fraction is the load-bearing quantity for a fixed installation sited near personnel: it is what lets the shield close at a few metres rather than the tens of tonnes a same-power D–T machine would demand (AG-L1-A2), and a burn-lean mode that pumps the bred triton before it burns strips a

Table 1. Aegis forward-base burner design point (gates BU-L1-A1, BU-L1-A10, BU-L2-A5). Every value is a frozen register anchor.

quantity	value	serial
engineering gain Q_E	1.318	BU-L1-A1
fusion power P_{fus}	4298.5 MW	BU-L1-A1
charged power P_{ch}	3868.0 MW	BU-L1-A10
neutron power P_n	233.95 MW	BU-L1-A1
neutron fraction f_n	5.44 %	BU-L1-A1
plug potential ϕ_i	248.75 keV	BU-L2-A5
carrier temperature T_e	89.7 keV	BU-L2-A5
ion temperature T_i	90 keV	BU-L1-A1
plug density ratio n_p/n_c	16	BU-L2-A5
mirror ratio R_m	4.61	BU-L1-A1
plugged confinement τ_p	3.73 s	BU-L1-A1
axial-loss power P_{ax}	2719.7 MW	BU-L1-A10
directed fraction f_{dir}	0.495	BU-L1-A10
central-cell field B_{0c}	5.50 T	BU-L1-A1
plug coil peak field	26.49 T	P3-FEA-10
net electric (islanded, forward-base)	+104 MW _e	AG-L1-A1
$P(Q_E > 1)$ over design ranges	1.0	KX-L1-A5-BU

Power flow of the Aegis forward-base burner (Schematic.)

**Figure 1.** Power flow of the Aegis forward-base burner (gates BU-L1-A1, BU-L1-A10). The primary D-³He channel delivers 3868.0 MW of charged power to the direct converter; the D-D side reactions contribute the 233.95 MW (5.44 %) neutron power; the bulk of the electric output recirculates to the plug and thermal barrier, leaving +104 MW_e net in the islanded forward-base configuration. **(Schematic.)**

further $\approx 60\%$ of the neutron power ($234 \rightarrow 93.5$ MW) for only $\approx 4\%$ fusion-power loss (AG-CX-16), an urban- and base-siting knob unique to the fuel.

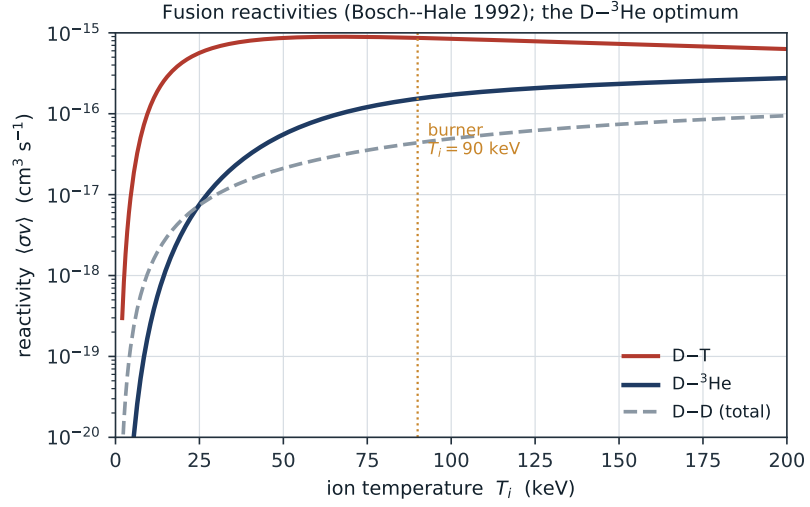


Figure 2. Maxwellian fusion reactivities from the Bosch-Hale parametrisation [9]. D- ^3He (navy) peaks an order of magnitude hotter than D-T (red) but is aneutronic in its primary channel; the burner operates at $T_i = 90$ keV (amber). The fuel choice is the origin of the 5.44% neutron fraction that makes a personnel-proximate forward-base installation possible.

4.3 The binding gate: the plug potential

Figure 3 plots the plug requirement of equation (6). Because the density ratio sits inside the exponent, the carrier temperature T_e is the dominant lever: the design point at $T_e = 89.7$ keV needs $n_p/n_c = 16$, but a modestly cooler carrier would drive the requirement up by an order of magnitude. This exponential sensitivity is why the synchrotron-pinned electron temperature of equation (11) is not a convenience but a prerequisite. Figure 4 shows the complementary Pareto structure (AG-CX-02): the plug potential saturates near 249 keV by a density ratio of ≈ 16 , while the engineering gain rides a steep cliff above it (Q_E climbing from 1.32 at $R = 16$ through 2.7 at $R = 20$ to 23 at $R = 24$). The canonical operating point $R = 16$ is deliberately placed at the *robust knee* of this curve, not in the fragile steep zone: it is the point at which the confining potential is essentially fully formed and the design is insensitive to the exact ratio, at the cost of forgoing the large but fragile gains available further up the cliff. This is a conservative choice, and it is the honest one for a system whose plug requirement is not yet demonstrated.

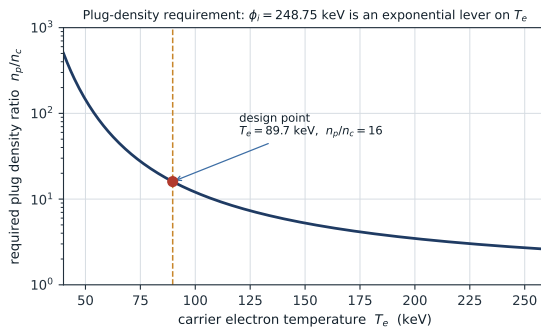


Figure 3. The plug-density requirement $n_p/n_c = \exp(e\phi_i/T_e)$ (gate BU-L2-A5). The design point sits at $n_p/n_c = 16$; the carrier temperature T_e is the exponential lever that sets it.

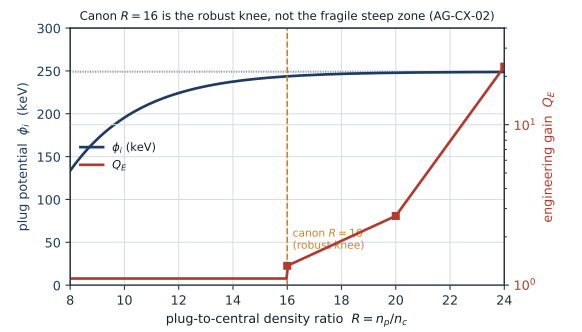


Figure 4. Plug-ratio Pareto (AG-CX-02): ϕ_i saturates near 249 keV by $R \approx 16$ while Q_E rides a steep cliff above it. Canon $R = 16$ is the robust knee, not the fragile steep zone.

4.4 The second lever: direct-converter efficiency and net-power break-even

Equation (8) is linear in η_{dec} , and figure 5 makes the consequence explicit. The converter module efficiency is a declared, load-bearing input on a conservative-to-ideal ladder: a conservative canon

floor $\eta_{\text{dec}} = 0.49$, a design basis $\eta_{\text{dec}} = 0.70$, and a proton-only ideal ceiling $\eta_{\text{dec}} = 0.85$ (BU-L1-A10, BU-SX-03). Net electric requires η_{dec} above a break-even near 0.52; the design point $Q_E = 1.318$ is reached at the design-basis efficiency, while the conservative floor sits *below* break-even and does not, on its own, deliver net electric. We state this plainly: the converter efficiency is a binding lever of the same class as the plug gate, not a free upside, and the design's net-power claim is explicitly conditional on the converter clearing the break-even — a hardware, not a design-study, result, treated in the direct-conversion companion [35]. The engineering-gain headline and the net-power headline are therefore reported with their conditions attached throughout.

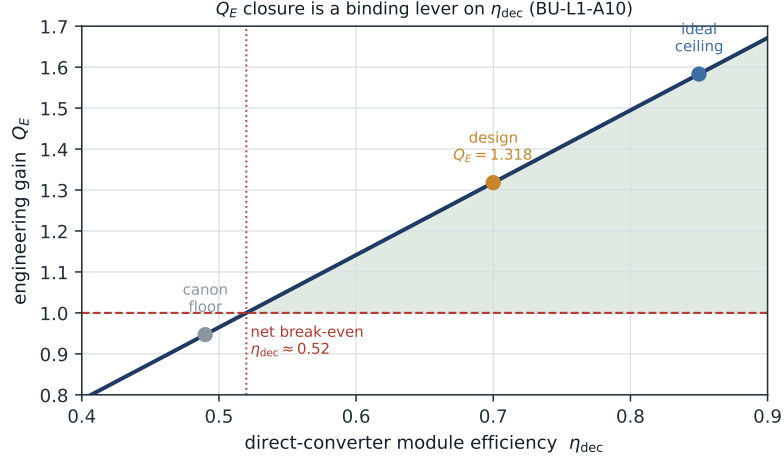


Figure 5. Engineering gain versus direct-converter efficiency (gate BU-L1-A10). Q_E is linear in η_{dec} ; net electric requires η_{dec} above the break-even near 0.52. The design point ($\eta_{\text{dec}} = 0.70$, $Q_E = 1.318$) closes; the conservative canon floor (0.49) does not. The converter efficiency is a binding lever, not a free upside.

4.5 Sensitivity and uncertainty

Which input actually controls the gain? Figure 6 gives the Saltelli–Sobol answer (HX-05): the plug density ratio is the first-order driver of Q_E , with a first-order index $S_1 = 0.41$ and a total-order index $S_T = 0.87$. The large gap $S_T - S_1 = 0.46$ is interaction variance, carried almost entirely with the central-cell beta: the plug gate and the central beta are the two quantities that jointly set the gain, and both are on the register's critical list. The corresponding Monte-Carlo distribution over the full design ranges (figure 7; KX-L1-A5-BU, 40 000 samples) is $Q_E = 1.317 \pm 0.14$ with $P(Q_E > 1) = 1.0$: the design closes robustly across its input ranges — conditional, as always, on the plug and converter levers being retired by experiment.

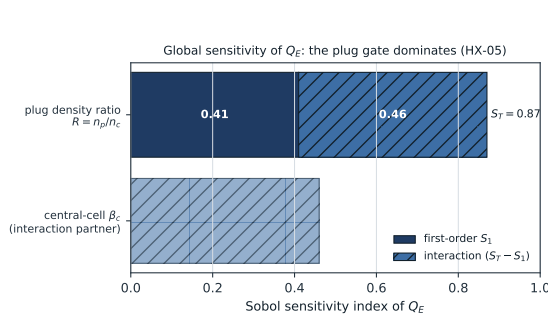


Figure 6. Global sensitivity of Q_E (Saltelli–Sobol, HX-05). The plug density ratio dominates: first-order $S_1 = 0.41$, total-order $S_T = 0.87$; the 0.46 interaction term is carried with the central-cell beta.

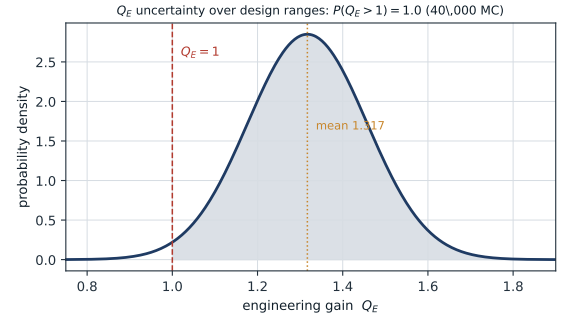


Figure 7. Monte-Carlo distribution of Q_E over the design ranges (KX-L1-A5-BU, 40 000 samples): $Q_E = 1.317 \pm 0.14$, $P(Q_E > 1) = 1.0$. Every value is conditional on the plug and converter gates.

4.6 Loss-cone microstability

Figure 8 shows the loss-cone microstability result reproduced independently with GPU PIC (AG-CX-03) [23]. The bare loss-cone distribution is unstable, with normalised growth rate

$\gamma/\omega_{ci} = 0.037$; a maintained energetic-stabiliser fill quenches it monotonically (0.024 at $f_s = 0.15$, 0.011 at 0.30, effectively suppressed by 0.45), consistent with the 6.25 % warm-plasma DCLC threshold of gate BU-L2-A5. The stabilisation is a threshold, not a margin: the plug is stable only while the warm-plasma fraction is actively held, which makes the warm-fill an operational control quantity handed to the burner control suite [43], and a weak residual AIC branch persists that the microstability companion [38] treats in full. Presenting this as a knee rather than a margin is part of the claim.

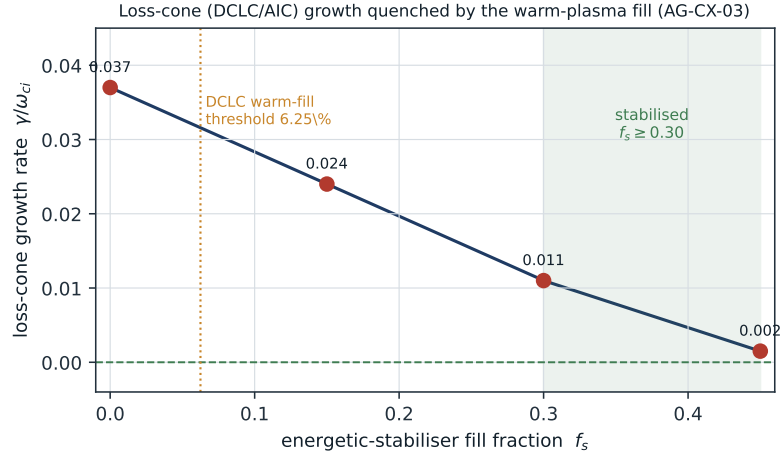


Figure 8. Loss-cone (DCLC/AIC) growth rate quenched by the energetic-stabiliser fill (AG-CX-03; GPU PIC). The bare distribution is unstable ($\gamma/\omega_{ci} = 0.037$) and is monotonically stabilised as the warm-plasma fraction is raised past the 6.25 % threshold. Stability is a maintained-control condition, not a passive margin.

4.7 Forward-base operating envelope

The physics core is the same for Aegis and MetroVolt; what makes the machine an *Aegis* is the forward-base operating envelope, which is where the product case is made or lost. Three quantities define it.

Black-start. A forward installation must start without the grid. Figure 9 shows the reduced black-start trajectory (HX-07, AG-CX-18): from a cold start the magnets charge, the plug potential builds, the central cell ignites, and the direct-converter output crosses the start-up draw (of order 340 MW_e) at ≈ 42 s on ~ 3 MWh of stored energy, after which the plant is grid-independent and settles to its +104 MW_e islanded steady state (AG-L1-A1). The superconducting-magnet charge is the long pole of the full cold-start sequence.

Islanded load-following. A defence site swings its directed-energy and radar loads by tens of megawatts. The direct converter trims on a millisecond timescale, and the ≈ 8 s plasma thermal lag is bridged by a small buffer of order 1 MWh (AG-CX-20), so the installation follows its own load without a grid tie; the plasma throttle is bounded below by burn stability, which the control-barrier safety clamp enforces [43].

Availability. Figure 10 gives the fleet capacity factor by redundancy tier from an event-driven Monte-Carlo (HX-01, AG-L1-A3): steady-state availability of 83.3/95.7/98.8 % across the tiers, with a P10–P90 capacity-factor spread of 68–78, 86–90 and 94–96 % respectively, under an $N = 4/2/2$ redundancy rule. The single-point-of-failure is the large superconducting magnet set, whose repair time dominates the availability even when everything else is redundant; coil demountability and quench protection are therefore the top reliability items [21], and they are a magnet-design, not a physics, problem [34].

4.8 The plug magnet and the fuel logistics

The plug coil is the high-field no-insulation REBCO magnet [19, 18, 20] distinct from the breeder centrepost; its conservative field basis is 26.49 T (versus the breeder centrepost’s 16.84 T — two separate magnets), and the three-dimensional FEA on as-built CAD closes with a reinforced allowable of 1091 MPa at that field, raising a thin-wall margin of $0.86\times$ to $1.34\times$ and holding the REBCO strain within a $2.85\text{--}3.54\times$ margin under a graded preload (P3-FEA-10) [34]. The current-sharing temperature at full current is 57.1 K, leaving a comfortable temperature margin at the operating point. On fuel, the forward-base logistics are trivial by design: the in-vessel helium-3

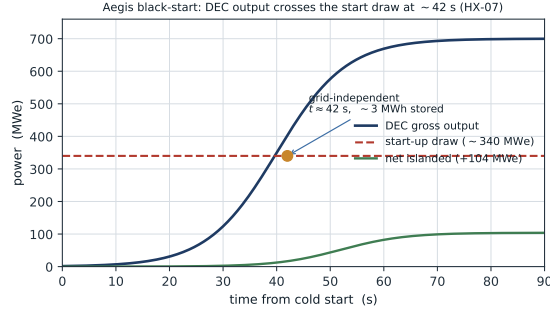


Figure 9. Aegis black-start (HX-07). The direct-converter output crosses the start-up draw at ≈ 42 s on ~ 3 MWh of stored energy; thereafter the installation is grid-independent at $+104$ MW_e islanded.

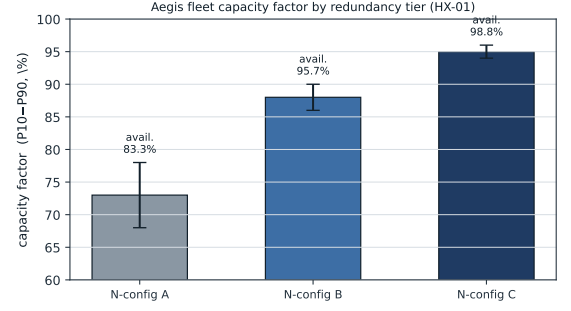


Figure 10. Aegis fleet capacity factor by redundancy tier (HX-01, $N = 4/2/2$). Bars show the P10–P90 spread; annotations give steady-state availability. The superconducting magnet set is the availability-limiting single point of failure.

start-up charge is ~ 0.3 g, a one-year grid-independent autonomy package is ≈ 21.3 kg of helium-3 plus ≈ 30 kg of deuterium (a ~ 1 -tonne package occupying under 0.8 m³), and there is no fuel convoy to interdict (AG-L1-A6, AG-L1-A8). The binding fuel constraint is not on-site but upstream: helium-3 at burner scale is lunar-gated, and the breeder is its strategic terrestrial co-producer at 37.66 kg-³He per full-power year per unit (AG-CX-17) [41, 32]. Table 2 collects the forward-base envelope.

Table 2. Aegis forward-base operating envelope. Serials refer to the register [30].

quantity	value	serial
islanded steady-state net	$+104$ MW _e	AG-L1-A1
black-start crossover	≈ 42 s / ~ 3 MWh	HX-07
start-up external draw	~ 340 MW _e	AG-L1-A1
load-following buffer	~ 1 MWh (ms DEC trim)	AG-CX-20
availability (redundancy tiers)	83.3/95.7/98.8 %	HX-01
capacity factor (P10–P90)	68–96 %	HX-01
1-year fuel package	21.3 kg ³ He + 30 kg D	AG-L1-A8
in-vessel ³ He start charge	~ 0.3 g	AG-CX-17
shield (fixed installation)	4.5 m / 20 kt class	AG-L1-A2
plug coil field basis	26.49 T	P3-FEA-10

4.9 Verification status

Table 3 states, for the load-bearing claims, whether each is robust (independent of the plug gate), conditional (contingent on retiring the plug and/or converter lever), or a declared input. The neutron-source and low-activation performance is robust; the net-power performance is conditional; the converter efficiency is a declared input on the ladder of figure 5. Naming the class of each claim is the honest way to present an advanced-fuel mirror.

Table 3. Evidence class of the load-bearing claims. “Robust” is independent of the plug gate; “conditional” is contingent on retiring the plug/converter levers by experiment.

claim	value	class
neutron fraction f_n	5.44 %	robust
neutron power P_n	233.95 MW	robust
fixed-installation shield	4.5 m class	robust
engineering gain Q_E	1.318	conditional (plug + η_{dec})
islanded net electric	$+104$ MW _e	conditional (plug + η_{dec})
plug density ratio n_p/n_c	16	requirement (open gate)
converter efficiency η_{dec}	$0.49/0.70/0.85$	declared input
loss-cone stability	$f_s \geq 6.25$ %	threshold (maintained)
plug-coil structure	$1.34\times$ margin	closed (FEA)

5 Discussion

5.1 Against the tandem-mirror record

The design’s confinement rests on a plug potential of 248.75 keV, and the honest comparison is to what the tandem-mirror programme actually demonstrated. GAMMA-10, TARA and MFTF-B

formed thermal-barrier and plug potentials of order a kilovolt and confirmed the tandem-mirror confinement principle [6, 5]; the Aegis requirement is more than two orders of magnitude beyond that record. We therefore do not claim the plug retired — we claim it is the binding gate, and that the whole design is structured to make that gate the single dominant risk (the Sobol result of figure 6 is the quantitative statement of exactly that). The modern high-temperature-superconductor mirror line [8, 7] is what makes a compact deep-plug device conceivable at all: a WHAM-class plug-potential experiment at reactor-relevant field is precisely the demonstration that would retire the gate, and it is the experiment against which our conditional claims should be read.

5.2 Against the direct-conversion and advanced-fuel literature

Direct energy conversion of a mirror end-loss stream is an old idea with a real experimental pedigree [12], and our treatment is deliberately conservative relative to it: we take the converter efficiency as a declared ladder (0.49/0.70/0.85) rather than an achieved number, and we show the net-power break-even near 0.52 explicitly (figure 5) rather than assuming it away. The advanced-fuel critiques of Rider and Nevins [11, 13] make the general point that non-thermal and advanced-fuel systems pay a steep recirculating-power penalty; our $Q_E = 1.318$ is a direct, quantitative acknowledgement of that penalty rather than a refutation of it — it is a low gain, honestly reported, and it is why the neutron-source and firm-power mission, not a high fusion gain, is the load-bearing value proposition of the machine [10]. The comparison to a compact high-field toroidal breeder [15, 16, 17] is complementary, not competitive: the breeder is the 14 MeV neutron source and helium-3 co-producer, and the burner is the low-neutron firm-power sink; the two are the two halves of one strategic-materials platform [32].

5.3 Why Aegis specifically

The forward-base case does not turn on a high gain — it turns on three properties a diesel or fission alternative cannot jointly offer: a sub-cubic-metre annual fuel charge with no convoy tail, a neutron output low enough to site near personnel behind a few metres of shield, and a black-start to grid-independent firm power in under a minute (figure 9). The islanded operating point (+104 MW_e) is modest but firm, and the availability (figure 10) is set by magnet demountability rather than by plasma physics. These are the properties that make the machine an Aegis rather than a laboratory burner, and they are why the product framing is forward-base defence power and not a merchant power plant. The economics of that mission are reported separately and are not part of this deposit.

6 Limitations

Four caveats bound the claims, each stated with its evidence class. First, every net-power number is contingent on retiring the plug-potential requirement $n_p/n_c = 16$, which a design study cannot do; only a plug-potential experiment can. Second, the converter efficiency is a declared input on a ladder, and the conservative canon floor (0.49) sits below the net-power break-even (≈ 0.52), so the net-power claim is conditional on the converter clearing that break-even in hardware. Third, the loss-cone stability is a maintained threshold at 6.25 % warm fill, not a passive margin, and a residual AIC branch persists. Fourth, the central-cell beta $\beta_c = 0.55$ enters the balance as an assumed input rather than an MHD-solved result, and the helium-3 fuel supply is a separate, lunar-gated problem [41]. None of these bears on the robust neutron-source and low-activation performance.

7 Conclusion

The Aegis forward-base burner reaches an engineering gain $Q_E = 1.318$ at 4298.5 MW fusion power with a 5.44 % neutron fraction, delivering firm, islanded, grid-independent power (+104 MW_e) with a black-start to grid-independence in ≈ 42 s and a fleet capacity factor of 68–96 %. The binding physics gate is the plug-potential requirement $n_p/n_c = \exp(e\phi_i/T_e) = 16$, retired only by a dedicated experiment and confirmed by the Sobol analysis as the first-order driver of the gain; the direct-converter efficiency is a second binding lever with a net-power break-even near 0.52; the loss-cone microstability is a maintained threshold at 6.25 % warm fill; and the plug coil qualifies structurally on as-built CAD. The neutron-source and low-activation performance — the properties that make a personnel-proximate, convoy-free forward-base installation possible — are robust and plug-independent; the net-power performance is conditional on the two levers. The design is presented with that separation carried openly, which is the only honest way to present an advanced-fuel mirror for a mission that will be fielded rather than merely published.

Acknowledgments

Part of the Kronos Fusion Energy 2026 design series. The authors thank the Kronos physics de-risking programme for the frozen result set underlying every quantitative claim, and the reviewers of the 2026 design series.

Reproducibility.

Every headline quantity in this paper is a frozen anchor (BU-L1-A1, BU-L1-A10, BU-L2-A5, KX-L1-A5-BU, AG-L1-A1, HX-01, HX-05, HX-07, P3-FEA-10) in the Kronos Physics De-Risking Register [30] (doi:10.5281/zenodo.22133057) and is regenerable from the deposited solver and the uniquely-named figure-generation script archived with this paper at doi:10.5281/zenodo.22645693; official version: <https://www.kronosfusionenergy.com/publications/aegis-burner.html>. The Bosch–Hale reactivities of figures 2 are the standard 1992 parametrisation [9]. No result depends on unpublished or proprietary data.

Scope — what this paper does not claim.

Q_E is the engineering gain at the frozen design point, not a demonstrated result; the plug density requirement is stated as the binding open gate, not solved; the converter efficiency is a declared input, not a measured value; and low-neutron is *not* aneutronic. No economic quantity is reported here.

Data availability

This paper is archived at doi:10.5281/zenodo.22645693. Headline quantities are regenerable from deposited code and cross the Kronos 2026 series: the breeder platform [32], the sibling MetroVolt burner [33], the REBCO magnet basis [34], direct energy conversion [35], the plug microstability [38] and RF alpha-channeling [37] companions, the ECH thermal-barrier and synchrotron papers [39, 40], the helium-3 supply analysis [41], and the islanded forward-base operations study [42]. Any economic analysis is reported separately and is not included in the public deposit.

Competing interests

Provisional patent applications relating to aspects of this work have been filed.

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